Chapter 14. Hermitian Forms

Sesquilinear forms and adjoint operators

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Hermitian forms on $\mathbb{C}$-vector spaces are Chapter 13 with a few bars (complex conjugations) added; the subtle differences justify a separate treatment. Proofs identical to those of Chapter 13 up to conjugation are only indicated. Throughout $F = \mathbb{C}$ and V is n-dimensional.

14.1 Hermitian forms

Motivation. Sesquilinear forms are linear in the first and conjugate linear in the second variable. They correspond to matrices via $H(v,w) = [v]^t J\overline{[w]}$, Hermitian ones to self-adjoint matrices, and their symmetry groups are the unitary groups $\{A : A^t J\bar{A} = J\}$.

Definition 14.1.1 (Sesquilinear and Hermitian forms)

$H : V \times V \to \mathbb{C}$ is **sesquilinear** if for all $u, v, w \in V$, $c \in \mathbb{C}$: (H1) $H(u+v,w) = H(u,w) + H(v,w)$, $H(cv,w) = cH(v,w)$; (H2) $H(u,v+w) = H(u,v) + H(u,w)$, $H(v,cw) = \bar{c}H(v,w)$. If moreover (H3) $H(v,w) = \overline{H(w,v)}$, H is a **Hermitian form** and (V,H) a **Hermitian space**.

Example 14.1.2

For a basis $\mathfrak{B} = \{v_i\}$, $[H]_\mathfrak{B} = (H(v_i,v_j))$ is the **matrix of** H, and for $v = \sum a_i v_i$, $w = \sum b_i v_i$:

$$H(v,w) = \sum_{i,j} a_i H(v_i,v_j)\bar{b}_j = [v]_\mathfrak{B}^t [H]_\mathfrak{B} \overline{[w]_\mathfrak{B}}.$$

Exercise 14.1.3

For $J \in \mathfrak{M}_{n,n}(\mathbb{C})$, $H_\mathfrak{B}^J(v,w) = [v]_\mathfrak{B}^t J\overline{[w]_\mathfrak{B}}$ is sesquilinear with $[H_\mathfrak{B}^J]_\mathfrak{B} = J$. (The dot product of $\mathbb{C}^n$ is $H_\mathcal{E}^I$.)

Solution. Linear in $[v]$, conjugate linear in $[w]$; $H_\mathfrak{B}^J(v_i,v_j) = e_i^t J e_j = J_{ij}$. □

Exercise 14.1.4

The sesquilinear forms $\mathfrak{Ses}(V)$ form a $\mathbb{C}$-vector space under pointwise operations.

Solution. As in Exercise 13.1.5. □

Theorem 14.1.5

$\Omega_\mathfrak{B} : \mathfrak{Ses}(V) \to \mathfrak{M}_{n,n}(\mathbb{C})$, $H \mapsto [H]_\mathfrak{B}$, is a vector space isomorphism.

Proof. Injective by Example 14.1.2, surjective by Exercise 14.1.3, linear entrywise. ■

Remark 14.1.6

With $\alpha_\mathfrak{B} : V \to \mathbb{C}^n$ and $J = [H]_\mathfrak{B}$, (V,H) and $(\mathbb{C}^n, H_\mathcal{E}^J)$ are essentially the same (the analogue of Remark 13.1.7).

Exercise 14.1.7

If $J \in \mathrm{GL}_n(\mathbb{C})$ and $A \in \mathfrak{M}_{n,n}(\mathbb{C})$, then $H^J_{\mathcal{E}}(AX, Y) = H^J_{\mathcal{E}}(X, (\bar{J}^{-1}A^*\bar{J})Y)$.

Solution. $(AX)^t J\bar{Y} = X^t A^t J\bar{Y}$, and $X^t J\,\overline{\bar{J}^{-1}A^*\bar{J}Y} = X^t JJ^{-1}\overline{A^*}J\bar{Y} = X^t A^t J\bar{Y}$, since $\overline{A^*} = A^t$. □

Exercise 14.1.8

A sesquilinear H is Hermitian iff $J = [H]_{\mathfrak{B}}$ satisfies $J^t = \bar{J}$, i.e. $J^* = J$. Such J are **self-adjoint** (**Hermitian**) matrices.

Solution. (H3) on basis vectors is $J_{ij} = \overline{J_{ji}}$. Conversely, by Example 14.1.2, $\overline{H(w,v)} = \overline{[w]^t J\overline{[v]}} = \overline{[w]}^t \bar{J}[v] = [v]^t \bar{J}^t[w] = [v]^t J[w]$ when $\bar{J}^t = J$. □

Remark 14.1.9

The Hermitian forms $\mathfrak{Her}(V)$ are not closed under multiplication by non-real scalars (iH is not Hermitian if $H \neq 0$), so $\mathfrak{Her}(V)$ is only a real vector space. Such subtle differences persist throughout Chapters 13 and 14.

Definition 14.1.10 (Unitary group)

$\mathbf{U}(V,H) = \{L \in \mathrm{GL}(V) : H(Lv, Lw) = H(v,w)\ \forall v, w\}$, the **unitary group**; its elements are **unitary operators**; this is **unitary geometry**.

Exercise 14.1.11

For a basis $\{v_i\}$, $\mathbf{U}(V,H) = \{L \in \mathrm{GL}(V) : H(Lv_i, Lv_j) = H(v_i, v_j)\ \forall i, j\}$.

Solution. By sesquilinearity, both sides of $H(Lv, Lw) = H(v,w)$ expand into the same combination of these values. □

Proposition 14.1.12

With $J = [H]_{\mathfrak{B}}$ (self-adjoint), $\{[L]^{\mathfrak{B}}_{\mathfrak{B}} : L \in \mathbf{U}(V,H)\} = \{A \in \mathrm{GL}_n(\mathbb{C}) : A^t J\bar{A} = J\}$.

Proof. As in Proposition 13.3.5: $H(Lv, Lw) = [v]^t([L]^t J\overline{[L]})\overline{[w]}$, which equals $[v]^t J\overline{[w]}$ for all v, w iff $[L]^t J\overline{[L]} = J$ (Exercise 9.3.7, applied with $\overline{[w]}$ ranging over $\mathbb{C}^n$). ■

This settles Remark 10.6.4 in the complex case ($J = I$).

Definition 14.1.13 (Matrix unitary groups)

For self-adjoint J, $\mathbf{U}^J_n(\mathbb{C}) = \{A \in \mathrm{GL}_n(\mathbb{C}) : A^t J\bar{A} = J\}$. (So $\mathbf{U}(n) = \mathbf{U}^I_n(\mathbb{C})$.)

Exercise 14.1.14

If J is invertible self-adjoint and $A \in \mathbf{U}^J_n(\mathbb{C})$, then $|\det A| = 1$.

Solution. $\det A \cdot \det J \cdot \overline{\det A} = \det J \neq 0$, so $|\det A|^2 = 1$. □

14.2 Non-degenerate Hermitian forms

Motivation. Orthogonality, non-degeneracy, the perp theorem and orthogonal bases transfer from quadratic spaces to Hermitian spaces, using the real and imaginary parts of H where Chapter 13 used polarization.

Definition 14.2.1 (Orthogonality)

$v \perp w$ if $H(v,w) = 0$; $S \perp T$ similarly; $S^\perp = \{v : H(w,v) = 0 \ \forall w \in S\}$. For Hermitian H, $H(v,w) = 0 \iff H(w,v) = \overline{H(v,w)} = 0$: orthogonality is symmetric.

Exercise 14.2.2

$S^\perp = \langle S \rangle^\perp \leq V$.

Solution. Conjugate linearity in the second variable still gives a subspace, and linearity in the first gives $S^\perp = \langle S \rangle^\perp$. □

Definition 14.2.3 (Orthogonal bases)

$\mathfrak{B}$ is an **orthogonal basis** if $[H]_\mathfrak{B}$ is diagonal and an **orthonormal basis** if $[H]_\mathfrak{B} = I$.

Definition 14.2.4 (Non-degenerate)

H is **non-degenerate** if $V^\perp = 0$. (E.g. the dot product $H^I_\mathcal{E}$ of $\mathbb{C}^n$.)

Proposition 14.2.5

H is non-degenerate iff $\det[H]_\mathfrak{B} \neq 0$.

Proof. As in the first proof of Proposition 13.5.10, replacing Y by $\bar{Y}$: H is degenerate iff some $\bar{Y} \neq 0$ has $X^t J \bar{Y} = 0$ for all X, iff $J\bar{Y} = 0$ for some $\bar{Y} \neq 0$. ■

Theorem 14.2.6 (Perp theorem, non-degenerate case)

If H is non-degenerate and $W \leq V$, then $\dim W^\perp = \dim V - \dim W$.

Proof. As in Theorem 13.5.11: with K the first m rows of J, $v \in W^\perp$ iff $K\overline{[v]} = 0$, i.e. $\overline{W^\perp}$ (in coordinates) is $\ker L_K$. By Exercise 10.3.10, $\dim W^\perp = \dim \ker L_K = n - m$ (rank and dimension theorems). ■

Exercise 14.2.7

For a non-degenerate Hermitian space and $U, W \leq V$: $(W^\perp)^\perp = W$, $(U+W)^\perp = U^\perp \cap W^\perp$, $U^\perp + W^\perp = (U \cap W)^\perp$.

Solution. As in Exercise 13.5.13, using Theorem 14.2.6. □

Exercise 14.2.8

If H is Hermitian and $H(w,w) \neq 0$, then $V = \langle w \rangle \oplus \langle w \rangle^{\perp}$.

Solution. $v = \frac{H(v,w)}{H(w,w)} w + \left(v - \frac{H(v,w)}{H(w,w)} w \right)$, and the second term lies in $\langle w \rangle^{\perp}$ by Hermitian symmetry ($H(w,\cdot) = \overline{H(\cdot,w)}$); $\langle w \rangle \cap \langle w \rangle^{\perp} = 0$. □

Observation 14.2.9

If H is Hermitian and $H(v,v) = 0$ for all v, then $H = 0$.

Proof. $H(v+w,v+w) - H(v,v) - H(w,w) = 2 \operatorname{Re} H(v,w)$, and as in the proof of Observation 10.6.5, $H(v+iw,v+iw) - H(v,v) - H(w,w) = 2 \operatorname{Im} H(v,w)$. Both vanish, so $H(v,w) = 0$. ■

Theorem 14.2.10

Every nonzero Hermitian space (V,H) has an orthogonal basis (H need not be non-degenerate).

Proof [Supplement]. As in Theorem 13.5.16: if $H = 0$ any basis works; otherwise pick w with $H(w,w) \neq 0$ (Observation 14.2.9), split $V = \langle w \rangle \oplus \langle w \rangle^{\perp}$ (Exercise 14.2.8) and induct on dimension. ■

14.3 H-identification and adjoint operators

Motivation. $v \mapsto H(\cdot, v)$ is now *conjugate* linear, but still a natural bijection $V \to V^*$ for non-degenerate H. Through it the dual map becomes the adjoint $L^\star$, with matrix $\bar{J}^{-1}[L]^* \bar{J}$, and unitary operators are those with $L^\star L = I$.

Remark 14.3.1

$\varphi_H^V : V \to V^*$, $\varphi_H^V(v)(w) = H(w,v)$, is well defined (linear in w), but it is *not* linear: H is conjugate linear in its second variable.

Exercise 14.3.2

$\varphi_H^V(u+v) = \varphi_H^V(u) + \varphi_H^V(v)$ and $\varphi_H^V(cv) = \bar{c}\varphi_H^V(v)$: φ_H^V is conjugate linear.

Solution. $H(w,u+v) = H(w,u) + H(w,v)$ and $H(w,cv) = \bar{c}H(w,v)$. □

Observation 14.3.3

For $W \leq V$: (a) $\varphi_H^V(W) \leq V^*$;
(b) if H is non-degenerate, $\dim W = \dim \varphi_H^V(W)$.

Proof. (a) Conjugate linear images of subspaces are subspaces ($c\varphi(w) = \varphi(\bar{c}w)$).

(b) If $\{w_j\}$ is a basis of W, then $\{\varphi(w_j)\}$ spans $\varphi(W)$. It is independent: $\sum c_j \varphi(w_j) = \varphi(\sum \bar{c}_j w_j) = 0$ forces $\sum \bar{c}_j w_j \in V^{\perp} = 0$ (as in Exercise 10.3.10). ■

Theorem 14.3.4

If H is non-degenerate, φ_H^V is a bijection (an isomorphism of abelian groups $(V,+)\to(V^*,+)$); every element of V^* is $\varphi_H^V(v)$ for a unique v.

Proof. Injective: $\varphi(u)=\varphi(v)$ gives $H(w,u-v)=0$ for all w, so $u=v$. Surjective: Observation 14.3.3(b) with $W=V$. ■

Since φ_H^V is not linear, it has no matrix, and there is no second proof of Proposition 14.2.5 via φ_H.

Notation 14.3.5 (H-identification)

For non-degenerate H, the basis-free bijection φ_H^V is used to identify $V=V^*$, $v=\varphi_v=\varphi_H^V(v)$. (This identification is only conjugate linear, and Remark 13.8.6 cannot be carried over.)

Exercise 14.3.6

If H is non-degenerate and $\{v_i\}$ is a basis, there is a basis $\{w_j\}$ with $H(v_i,w_j)=\delta_{ij}$.

Solution. $w_j=(\varphi_H^V)^{-1}(v_j^*)$; these form a basis by Observation 14.3.3(b). □

Exercise 14.3.7

If V has an orthonormal basis $\{v_i\}$, then $\varphi_H^V(v_i)=v_i^*$.

Solution. $\varphi_H^V(v_i)(v_k)=H(v_k,v_i)=\delta_{ki}$. □

Exercise 14.3.8

If H is non-degenerate, the square

$$\begin{array}{ccc} V\times V & \xrightarrow{H} & \mathbb{C} \\ {\scriptstyle I\times\varphi_H^V}\downarrow & & \downarrow{\scriptstyle I} \\ V\times V^* & \xrightarrow[\varepsilon]{} & \mathbb{C} \end{array}$$

commutes: under H-identification, $H=\varepsilon$.

Solution. $\varepsilon(v,\varphi_H^V w)=\varphi_H^V(w)(v)=H(v,w)$. □

Theorem 14.3.9 (Perp theorem, non-degenerate case)

If H is non-degenerate and $W\le V$, then under H-identification $W^\perp=W^{\mathrm{perp}}$; in particular $\dim W^\perp=\dim V-\dim W$.

Proof. By Exercise 14.3.8, $v\in W^\perp \iff \varphi_v\in W^{\mathrm{perp}}$, i.e. $\varphi_H^V(W^\perp)=W^{\mathrm{perp}}$. By Observation 14.3.3(b) and Theorem 13.7.3, $\dim W^\perp=\dim W^{\mathrm{perp}}=\dim V-\dim W$. ■

From now on (V,H) and (W,K) are non-degenerate Hermitian spaces with bases $\mathfrak{B},\mathfrak{C}$.

Definition 14.3.10 (Adjoint map)

(a) For $L \in \mathfrak{L}(V,W)$, the **adjoint map** $L^\star \in \mathfrak{L}(W,V)$ (with respect to H, K) is L^* under H- and K-identification:

$$\begin{array}{ccc} W & \xrightarrow{L^\star} & V \\ {\scriptstyle \varphi_K^W}\downarrow & & \downarrow{\scriptstyle \varphi_H^V} \\ W^* & \xrightarrow[L^*]{} & V^* \end{array}$$

i.e. $L^\star = (\varphi_H^V)^{-1} \circ L^* \circ \varphi_K^W$, which is linear (a composite of two conjugate linear maps and a linear one).
(b) For $L \in \mathfrak{L}(V,V)$, the **adjoint operator** is $L^\star = (\varphi_H^V)^{-1} \circ L^* \circ \varphi_H^V$.

Proposition 14.3.11

$L^\star$ is the unique linear map with $K(Lv,w) = H(v,L^\star w)$ for all $v \in V$, $w \in W$ (equivalently $K(w,Lv) = H(L^\star w, v)$).

Proof. As in Proposition 13.9.3: $K(Lv,w) = \varphi_w(Lv) = (L^*\varphi_w)(v) = \varphi_{L^\star w}(v) = H(v,L^\star w)$; uniqueness by non-degeneracy. The second form follows by conjugating. ■

Proposition 14.3.12

(a) With $J_{\mathfrak{B}} = [H]_{\mathfrak{B}}$, $J_{\mathfrak{C}} = [K]_{\mathfrak{C}}$: $[L^\star]_{\mathfrak{C}}^{\mathfrak{B}} = \overline{J_{\mathfrak{B}}}^{-1}([L]_{\mathfrak{B}}^{\mathfrak{C}})^* \overline{J_{\mathfrak{C}}}$. (b) For $L \in \mathfrak{L}(V,V)$ and $J = [H]_{\mathfrak{B}}$: $[L^\star]_{\mathfrak{B}}^{\mathfrak{B}} = \bar{J}^{-1}([L]_{\mathfrak{B}}^{\mathfrak{B}})^* \bar{J}$.

Proof. φ_H, φ_K are not linear, so we use Proposition 14.3.11. It reads $([Lv]_{\mathfrak{C}})^t J_{\mathfrak{C}} \overline{[w]_{\mathfrak{C}}} = [v]_{\mathfrak{B}}^t J_{\mathfrak{B}} \overline{[L^\star w]_{\mathfrak{B}}}$, i.e. $[v]^t([L]_{\mathfrak{B}}^{\mathfrak{C}})^t J_{\mathfrak{C}} \overline{[w]} = [v]^t J_{\mathfrak{B}} \overline{[L^\star]_{\mathfrak{C}}^{\mathfrak{B}}}\, \overline{[w]}$ for all v, w. By Exercise 9.3.7, $([L]_{\mathfrak{B}}^{\mathfrak{C}})^t J_{\mathfrak{C}} = J_{\mathfrak{B}} \overline{[L^\star]_{\mathfrak{C}}^{\mathfrak{B}}}$; conjugate and solve. (Without the bars this is a second proof of Proposition 13.9.4.) ■

Exercise 14.3.13

If $\mathfrak{B}, \mathfrak{C}$ are orthonormal, $[L^\star]_{\mathfrak{C}}^{\mathfrak{B}} = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^*$.

Solution. $J_{\mathfrak{B}} = J_{\mathfrak{C}} = I$ in Proposition 14.3.12. □

Proposition 14.3.14

$\mathbf{U}(V,H) = \{L \in \mathrm{GL}(V) : L^\star \circ L = I\}$.

Proof. $H(Lv,Lw) = H(v,L^\star Lw)$ equals $H(v,w)$ for all v, w iff $L^\star L = I$ (non-degeneracy, as in Observation 13.9.2). ■

Observation 14.3.15

For $L, M \in \mathfrak{L}(V,W)$, $c \in \mathbb{C}$: (a) $(L+M)^\star = L^\star + M^\star$;
(b) $(cL)^\star = \bar{c}L^\star$.

Proof. (b) $(cL)^\star = (\varphi_H^V)^{-1}(cL^*)\varphi_K^W = \bar{c}(\varphi_H^V)^{-1}L^*\varphi_K^W$, since $(\varphi_H^V)^{-1}$ is conjugate linear. (a) is similar (or use Proposition 14.3.11). Here "$\star = *$" would give the false $(cL)^\star = cL^\star$. ■

Exercise 14.3.16

For $L \in \mathfrak{L}(V,W)$, $M \in \mathfrak{L}(W,U)$: (a) $I^\star = I$;
(b) $(M \circ L)^\star = L^\star \circ M^\star$;
(c) if L is invertible, $(L^\star)^{-1} = (L^{-1})^\star$;
(d) $(L^\star)^\star = L$.

Solution. Use Proposition 14.3.11:

(b) $U(MLv,u) = K(Lv,M^\star u) = H(v,L^\star M^\star u)$.

(d) $K(Lv,w) = H(v,L^\star w) = \overline{H(L^\star w,v)} = \overline{K(w,(L^\star)^\star v)} = K((L^\star)^\star v,w)$. (a), (c) follow. □

Exercise 14.3.17

For $L \in \mathfrak{L}(V,V)$: (a) $\mathrm{tr} L^\star = \overline{\mathrm{tr} L}$;
(b) $\det L^\star = \overline{\det L}$;
(c) $\phi_{L^\star}(t) = \overline{\phi_L}(t)$ (conjugate coefficients);
(d) $m_{L^\star} = \overline{m_L}$.

Solution. $[L^\star] = \bar{J}^{-1}[L]^*\bar{J}$ is similar to $[L]^* = \overline{[L]}^t$, whose trace, determinant and characteristic polynomial are the conjugates of those of $[L]$.

(d) $f(L^\star) = (\bar{f}(L))^\star$ by Observation 14.3.15 and Exercise 14.3.16. □

Exercise 14.3.18

For $L \in \mathfrak{L}(V,W)$: (a) $\ker L^\star = (\mathrm{im}\, L)^\perp$;
(b) $\mathrm{im}\, L^\star = (\ker L)^\perp$.

Solution. As in Exercise 13.9.13, using Proposition 14.3.11, Exercise 14.2.7 and $(L^\star)^\star = L$. □